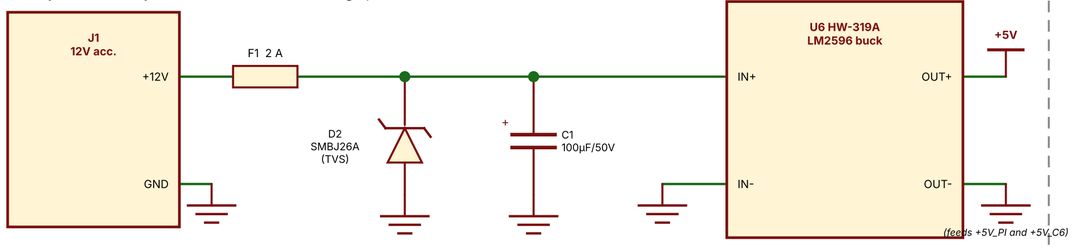
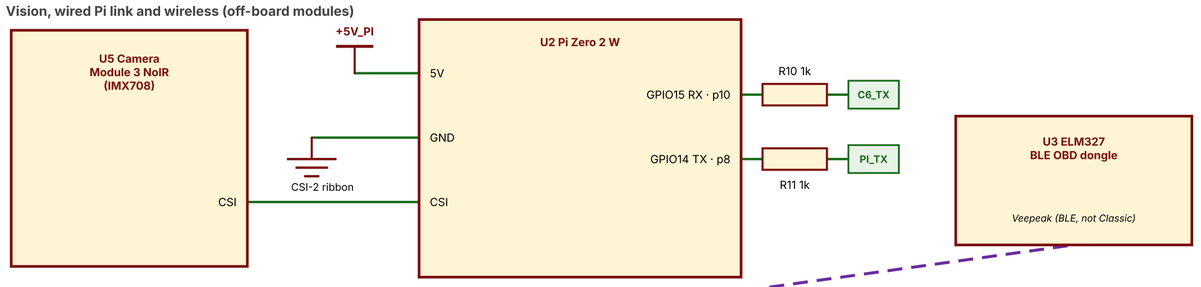
DRIFTS — Schematic (NewBrain architecture) · rev 8

Team 1 · ESP32-C6 dev kit + Raspberry Pi Zero 2 W · rev 8: split 5 V supply (Pi and ESP32 each fed from the charger), 850 nm IR board on its own port, Camera Module 3 NoIR · UART Pi link (rev 7) · Whisplay HAT · generated drawing (not a KiCad file)

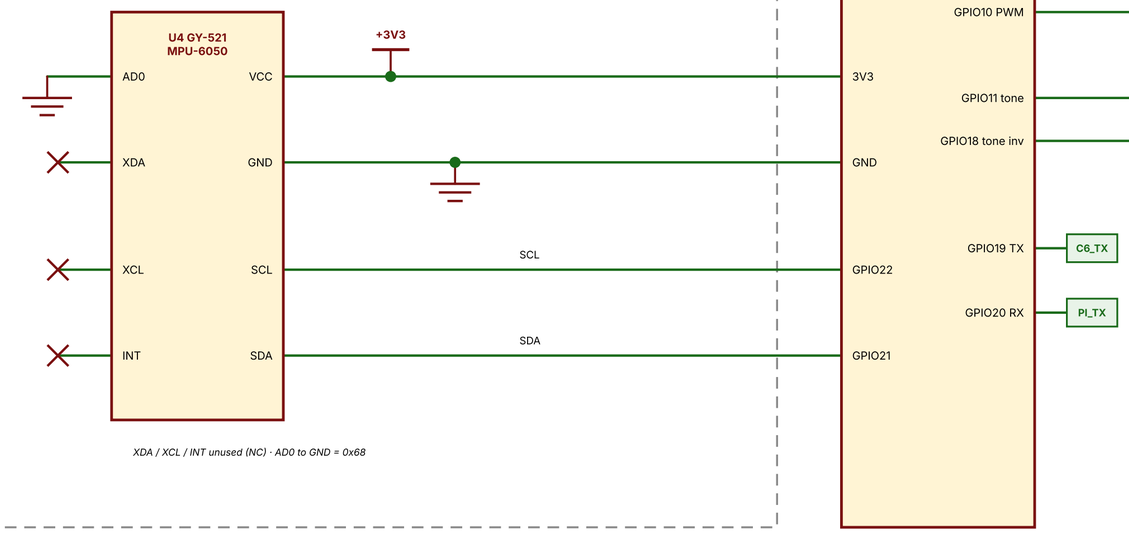
Power input: raw 12 V option (the car build uses a charger, rev 8 block)



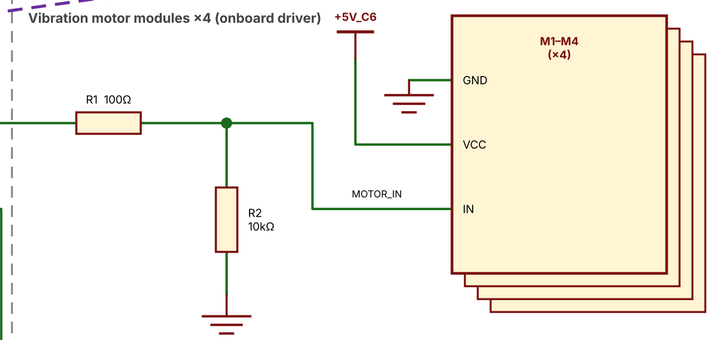
Vision, wired Pi link and wireless (off-board modules)



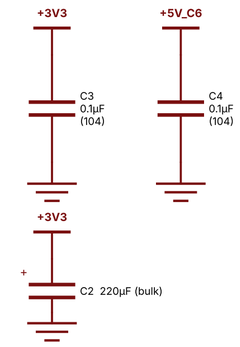
I2C bus · MPU-6050 (100 kHz)



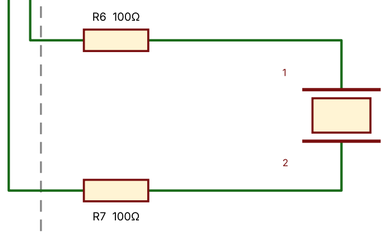
Vibration motor modules ×4 (onboard driver)



Decoupling



Piezo buzzer (differential drive)



Off = both pins LOW (0 V across LS1), never duty 0. Q2/R3/R4 of rev 2 removed.

Driver screen · PiSugar Whisplay HAT on U2 (40-pin header, no wires)

U12 PiSugar Whisplay HAT
stacked on U2 Pi Zero 2 W

LCD 1.69in 240×280 · SPI0

RGB LED · GPIO 25/24/23

Button · GPIO 17

ES8389 audio · I2S + I2C1

ST7789P3 · SPI0 MOSI 10, SCLK 11, CE0 8

DC 27 · RST 4 · backlight 22 (LOW = on)

RGB LED LOW = lit · button HIGH = pressed

I2S 18/19/20/21 · I2C1 2/3 · 5V/GND from U2 (ES8389 codec)

Speaker: onboard, or PH2.0 external (switch)

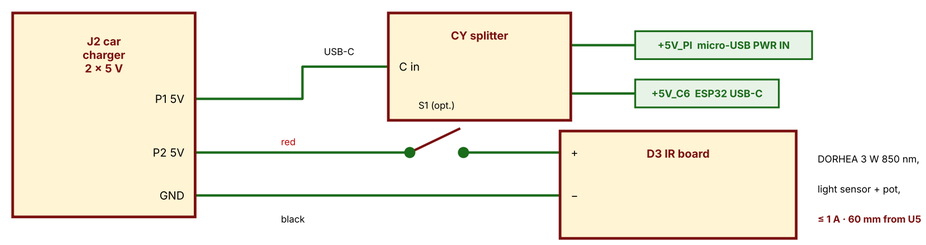
Shows U1's 1 Hz status (UART to U2): score,

level, links. Voice: Voice/clips via aplay.

Software: Raspberry/drifts/whisplay_ui.py

Details: Hardware/whisplay-hat.md

Car power (rev 8): 2-port charger, split supply, IR board on its own port



NOTES

1. U6 = HW-319A-V7.0.1 LM2596 module. Trim the output to 5.0 V with the trimmer BEFORE connecting the Pi or C6. Bench/demo: skip the 12 V front end and use a USB power bank.
2. Motor modules M1–M4 (GND, VCC, IN) have an onboard driver. IN ← GPIO10 (20 kHz PWM, 8-bit) via R1 100Ω; R2 10k holds IN low during reset.
3. Module VCC from +5V (the C6's 3.3 V LDO can't also start 4 motors). 3 V motors: lower DROWSY_VIB_DUTY (~150). C1 near the modules.
4. U4 GY-521: AD0→GND = 0x68. Wire.begin(21, 22). Breakout has pull-ups.
5. I2C runs at 100 kHz and carries only U4 (3.3 V: no level shifter). I2C SCAN lists 0x68. Free C6 pins: 2, 3, 23.
6. Status light: the Whisplay HAT's RGB LED (U12). U1 has no LEDs; GPIO 2, 3, 23 are free.
7. LS1 passive piezo, differential: GPIO11→R6→LS1←R7←GPIO18, one LEDC channel, GPIO18 inverted; 5 kHz 50 %, 6.6 Vpp. Off = both pins LOW.
8. C3/C4: 0.1µF ceramic at +3V3 and +5V near U1; one per module VCC.
9. Wireless (dashed purple): OBD dongle↔C6 (and the phone) over BLE. U3 must be BLE, not Classic. The Pi is wired: note 15. Camera U5 → Pi only.
10. Do not use: 4, 5, 8, 9, 15 (strapping); 6, 7 (JTAG); 12, 13 (USB); 16, 17 (UART0 to CH343); 14, 24–30 (flash).
11. Same-named flags and tags (+5V, GND, C6_TX, PI_TX) are one net.
12. Firmware v21: the 1 Hz status JSON goes to U2 over the UART and to a phone over BLE. See Hardware/pinout.md.
13. U12 Whisplay HAT: screen, RGB LED, speaker, button, driven by U2. The alert (buzzer, motors) stays on U1 and works without U2.
14. Power (rev 8): charger port 1 → splitter → U2 (+5V_PI) and U1 + motors (+5V_C6); port 2 → D3. Pi pin 2/4 no longer to U1; GND pin 6 kept for the UART. Bench: one supply via U2 pin 2/4 (5c). U6: raw 12 V.
15. Pi link (rev 7): UART1 115200 8N1, firmware v21. C6 GPIO19 TX → R10 1k → Pi GPIO15 RX (pin 10); Pi GPIO14 TX (pin 8) → R11 1k → C6 GPIO20 RX; GND to Pi pin 6. Both 3.3 V: no level shifter. Twist the 3 wires together, away from the motor and piezo wires.
16. D3 IR board (rev 8): 850 nm, up to ~1 A at 5 V, on charger port 2 only (never U2's or U1's pins). Mount 60 mm from U5's lens with nylon screws; heatsink on its back; S1 optional. schematic.md 5d.